



St Aloysius' College
Year 12 Mid-Year Examinations
2010

EXTENSION 2 MATHEMATICS
(Additional paper)

Total marks: 75

Reading time - 5 minutes

Working time – 2 hours

Examination papers must NOT be removed
from the examination room.

General Instructions

- Start each question in a new booklet
- Board approved calculators may be used
- Marks may be deducted for careless or badly arranged work
- Show all necessary work

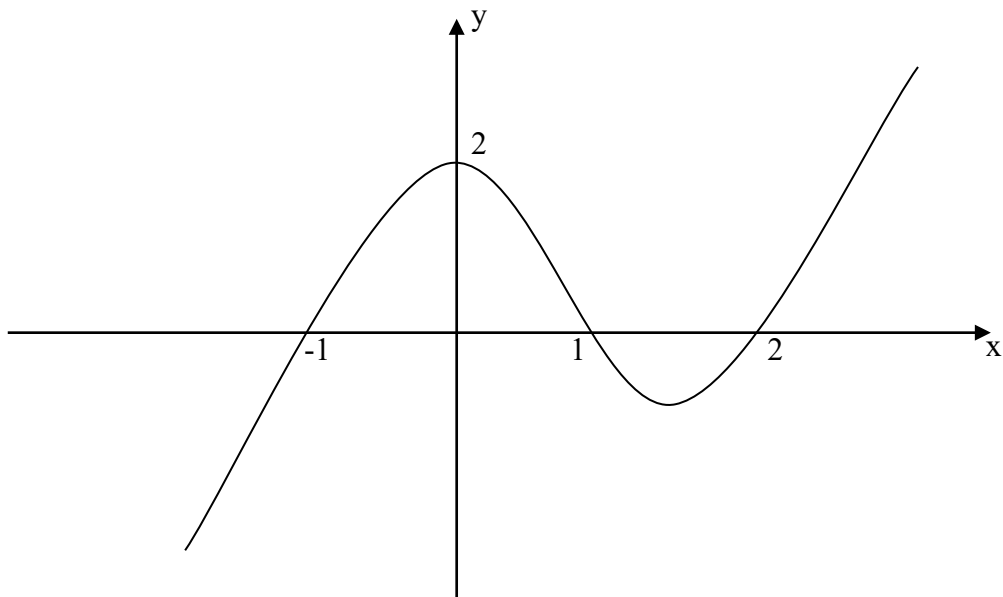
SUPERVISOR'S INSTRUCTIONS:

Please issue four 8 page answer booklets.

Please collect the examination paper with the answer booklets.

Question 1 (20 marks)

(a)



Use the graph of $y = f(x)$ shown above to make half-page sketches of the following.

(i) $y = \frac{1}{f(x)}$ (3)

(ii) $y = f(|x|)$ (3)

(iii) $y^2 = f(x)$ (3)

(iv) $y = |f(x)|$ (3)

(v) $y = \ln f(x)$ (3)

(b) (i) Show that $3x - 5 + \frac{4}{x+1} = \frac{3x^2 - 2x - 1}{x+1}$ (1)

(ii) State the equations of the vertical and oblique asymptotes of the function $y = \frac{3x^2 - 2x - 1}{x+1}$ (2)

(iii) Sketch the function showing the above features. (2)
[Note that you are **not** required to find the coordinates of any stationary points.]

START A NEW BOOKLET**Question 2** (20 marks)

- (a) $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which has eccentricity e foci S and S' .

(i) Show that the normal at P is $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 e^2$ (3)

(ii) The normal at P intersects the x axis at G .
Show that $PG^2 = a^2(e^2 - 1)(e^2 \sec^2 \theta - 1)$ (4)

(iii) Show that $PG^2 = (e^2 - 1)PS \times PS'$ (3)

- (b) $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ are distinct points on $xy = 4$.

M is the midpoint of the interval PQ , and T is the point of intersection of the tangents at P and Q .

Given that the tangent at $(2t, \frac{2}{t})$ is $x + t^2 y = 4t$

(i) Show that T has coordinates $\left(\frac{4pq}{p+q}, \frac{4}{p+q} \right)$ (2)

(ii) If M always lies on the line $x = 3$, find the locus of T including any restrictions. (2)

- (c) $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b$. Let M be the midpoint of the chord PQ , and O be the origin.

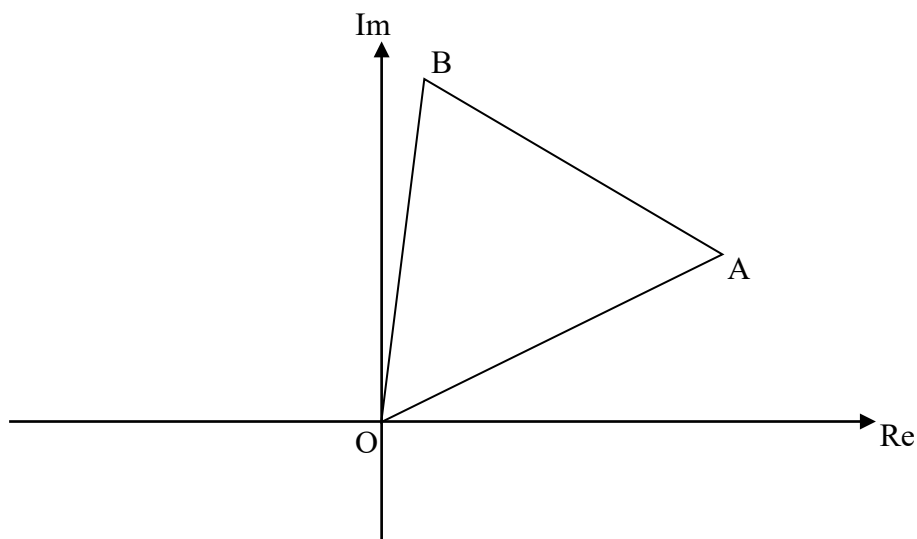
(i) Show that the product of the gradients of OM and PQ is $-\frac{b^2}{a^2}$. (3)

(ii) Given that $\sin(\theta + \phi)\sin(\theta - \phi) = \cos^2 \phi - \cos^2 \theta$, show that the acute angle, α , between OM and PQ is given by (3)

$$\tan \alpha = \frac{2ab}{a^2 - b^2} |\operatorname{cosec}(\theta + \phi)|$$

START A NEW BOOKLET**Question 3** (15 marks)

- (a) Solve the equation $(3-i)z + (2+3i) = \overline{8+7i}$ for the complex number z . (3)
- (b) Shade the region on the Argand diagram in which $|z-i| \leq \operatorname{Im} z$ and $|z| \leq 1$ both hold. (3)
- (c) Find $(-\sqrt{3}+i)^7$ in the form $a+ib$ (3)
- (d) One root of $z^2 + (3-2i)z + k = 0$ is $-4+2i$
- (i) Find the other root, and (1)
- (ii) Find the value of k . (1)
- (e) The Argand diagram below shows an equilateral triangle OAB .
A complex number z is represented by the vector OA . Let $\omega = \frac{-1+i\sqrt{3}}{2}$



Express the complex numbers represented by AB and OB in terms of z and ω (2)

- (f) Find the Cartesian equation of the locus of z if $|z^2 + (\operatorname{Re} z)^2| = 4$ (2)

START A NEW BOOKLET**Question 4** (20 marks)

- (a) A sequence is defined by $T_1 = 1$, $T_2 = 4$ and $T_n = 6T_{n-2} + T_{n-1}$ for integers $n > 2$. (5)

Use induction to prove that $T_n = \frac{2 \times 3^n - (-2)^{n-1}}{5}$ for $n \geq 1$

- (b) (i) Show that $x + y \geq 2\sqrt{xy}$ for all positive real numbers x and y . (1)

- (ii) Hence, or otherwise, show that $\log_2(2^n + 2^m) \geq \frac{n+m+2}{2}$ for all real n and m . (3)

- (c) (i) Use induction to prove that $(a+1)^n - a^n > na^{n-1}$ for real $a > 0$ and integers $n \geq 2$. (3)

In the following you may use the similar identity $(a+1)^n - a^n < n(a+1)^{n-1}$ without proof.

- (ii) Hence deduce that $a < \sqrt[n]{\frac{(a+1)^n - a^n}{n}} < a+1$ (1)

- (iii) (α) The straight line $y = [(a+1)^n - a^n]x + a^n - [(a+1)^n - a^n]$ intersects the curve $y = x^n$ at $x = a$ and $x = a+1$ where $a > 0$. (3)

Using calculus and part (ii), show that the function

$$f(x) = [(a+1)^n - a^n]x + a^n - [(a+1)^n - a^n] - x^n$$

has a maximum stationary value in the interval $a < x < a+1$

- (β) Deduce that $f(x) > 0$ for $a < x < a+1$ (1)

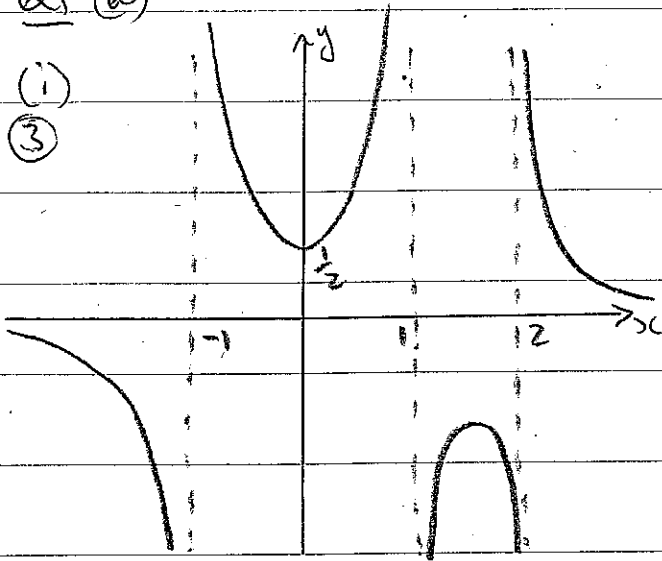
- (iv) Using the above results and by comparing the area under $y = x^n$ for $0 \leq x \leq n$ where $n \geq 2$ with an approximation for that area, show that

$$\frac{n^{n+1}}{n+1} < 1 + 2^n + 3^n + \dots + (n-1)^n + \frac{n^n}{2} \quad \text{for integers } n \geq 2 \quad (3)$$

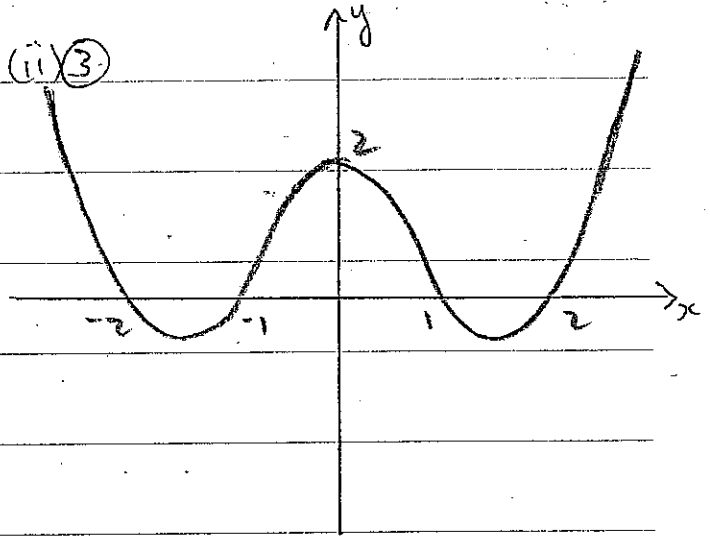
E_v+2

Q1 (a)

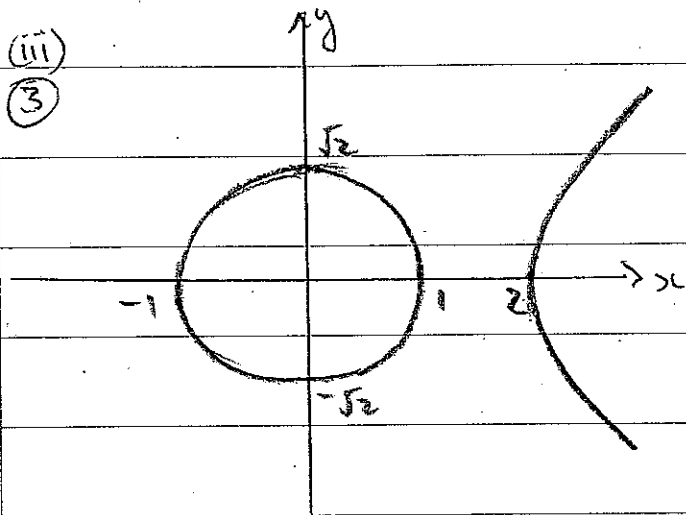
(i)
(3)



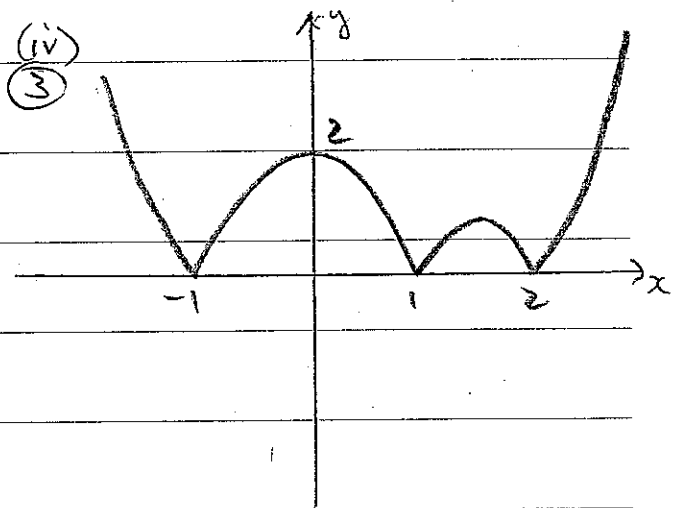
(ii) (3)



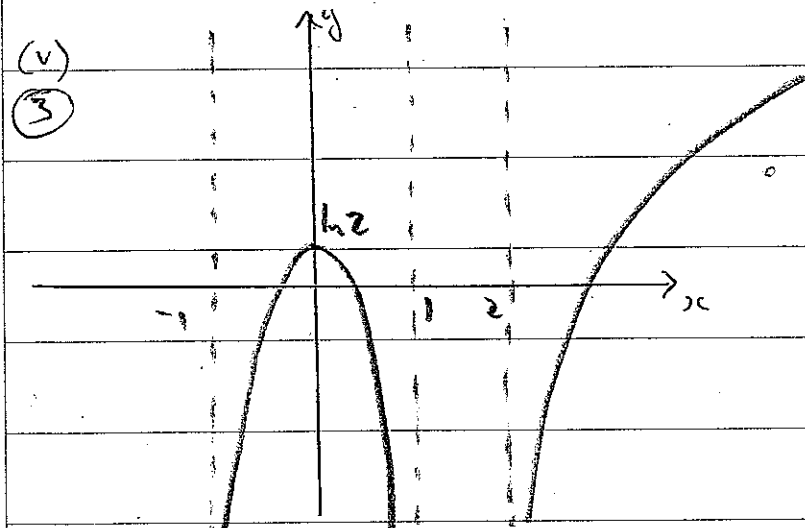
(iii)
(3)



(iv)
(3)

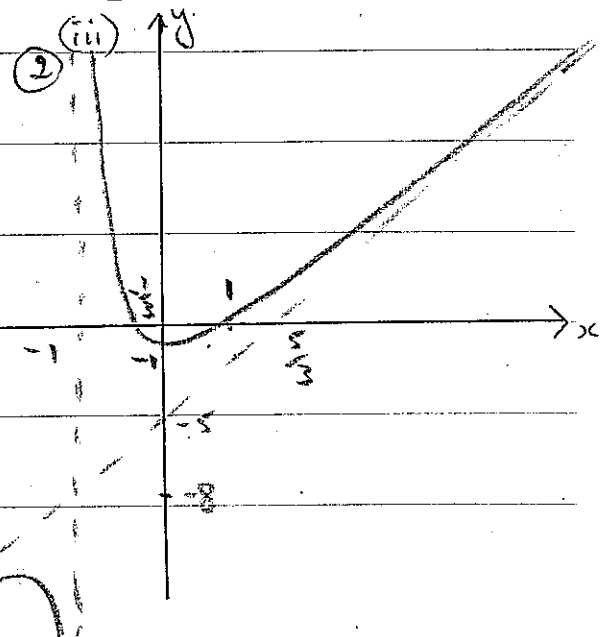


(v)
(3)



$$\begin{aligned} \text{(i)} \quad \frac{3x-5}{x+1} + \frac{4}{x+1} &= \frac{(3x-5)(x+1) + 4}{x+1} \\ &= \frac{3x^2 - 2x - 1}{x+1} \end{aligned}$$

$$\text{(ii)} \quad x = -1 \text{ and } y = 3x - 5$$



$$Q2 \ a) \ (i) \ \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b \sec^2 \theta}{a \sec \theta \tan \theta} = \frac{b \sec \theta}{a \tan \theta}$$

$$\therefore \text{Normal } y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$\frac{by}{\tan \theta} - b^2 = -\frac{ax}{\sec \theta} + a^2$$

$$\therefore \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 = a^2 e^2$$

$$b^2 = a^2 (e^2 - 1)$$

$$\therefore a^2 + b^2 = a^2 e^2$$

$$(ii) \ G(a e^2 \sec \theta, 0)$$

$$PG^2 = (a e^2 \sec \theta - a \sec \theta)^2 + (b \tan \theta)^2$$

$$= a^2 \sec^2 \theta (e^2 - 1)^2 + a^2 (e^2 - 1) \tan^2 \theta$$

$$= a^2 (e^2 - 1) (\cancel{\sec^2 \theta} + \cancel{\tan^2 \theta}) ((e^2 - 1) \sec^2 \theta + \tan^2 \theta)$$

$$= a^2 (e^2 - 1) ((\cancel{e^2 - 1}) \sec^2 \theta + \cancel{\sec^2 \theta} - 1)$$

$$= a^2 (e^2 - 1) (e^2 \sec^2 \theta - 1)$$

$$(iii) \ PS \times PS' = e \cdot PN \times e \cdot PN'$$

$$= e^2 \left(a \sec \theta - \frac{a}{e} \right) \left(a \sec \theta + \frac{a}{e} \right)$$

$$= a^2 e^2 \left(\sec^2 \theta - \frac{1}{e^2} \right)$$

$$= a^2 (e^2 \sec^2 \theta - 1)$$

$$\therefore (e^2 - 1) PS \times PS' = a^2 (e^2 - 1) (e^2 \sec^2 \theta - 1)$$

$$= PG^2$$

(b) $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ $xy = 4$

(i) Tangents $\begin{cases} x + p^2 y = 4p \\ x + q^2 y = 4q \end{cases} \Rightarrow (p^2 - q^2)y = 4(p - q)$ as $p \neq q$
 $\therefore y = \frac{4}{p+q}$
 $\therefore x = 4p - \frac{4p^2}{p+q} = \frac{4pq}{p+q} \therefore T\left(\frac{4pq}{p+q}, \frac{4}{p+q}\right)$ 2

(ii) $M\left(p+q, \frac{pq}{p+q}\right)$ lies on $x = 3 \therefore p+q = 3$

$\therefore T\left(\frac{4pq}{3}, \frac{4}{3}\right)$ 1 + 1

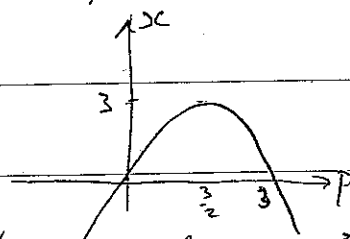
$\therefore$ Locus is $y = \frac{4}{3}$ for $x < 3$ but $x \neq 0$ *

* Restrictions: First $p \neq 0$ and $q \neq 0 \therefore x \neq 0$

Second, whether P and Q are on different branches or both on the 1st quadrant branch, T can never lie above that branch. $\Rightarrow x < 3$

OR for the second part: $p+q = 3 \Rightarrow q = 3-p$

$\therefore$ For T $x = \frac{4p(3-p)}{3}$



$\therefore x \leq 3$ with equality only if $p = q = \frac{3}{2}$

but $p \neq q \therefore x < 3$

The $p=3$ case is interesting as then $q=0 \therefore$ The tangent at Q "is" the y axis and the tangent at P has y intercept $(0, \frac{4}{3})$. But of course $q \neq 0$, so once again $x \neq 0$.

$$(c) (i) M \left(\frac{a(\cos\theta + \cos\phi)}{2}, \frac{b(\sin\theta + \sin\phi)}{2} \right)$$

$$\therefore M_{on} = \frac{b}{a} \frac{\sin\theta + \sin\phi}{\cos\theta + \cos\phi}$$

$$M_{pa} = \frac{b}{a} \frac{\sin\theta - \sin\phi}{\cos\theta - \cos\phi}$$

$$\therefore M_{on} \cdot M_{pa} = \frac{b^2}{a^2} \frac{\sin^2\theta - \sin^2\phi}{\cos^2\theta - \cos^2\phi}$$

$$= \frac{b^2}{a^2} \frac{\sin^2\theta - \sin^2\phi}{(1 - \sin^2\theta) - (1 - \sin^2\phi)} = -\frac{b^2}{a^2}$$

$$(ii) \tan\alpha = \left| \frac{\frac{b}{a} \left(\frac{\sin\theta + \sin\phi}{\cos\theta + \cos\phi} - \frac{\sin\theta - \sin\phi}{\cos\theta - \cos\phi} \right)}{1 + \left(-\frac{b^2}{a^2} \right)} \right|$$

$$= \frac{ab}{|a^2 - b^2|} \left| \frac{\cancel{\sin\theta\cos\theta} + \cancel{\sin\phi\cos\theta} - \cancel{\sin\theta\cos\theta} - \cancel{\sin\phi\cos\theta} - \cancel{\sin\theta\cos\theta} - \cancel{\sin\phi\cos\theta} + \cancel{\sin\phi\cos\theta} + \cancel{\sin\theta\cos\theta}}{\cos^2\theta - \cos^2\phi} \right|$$

$$= \frac{ab}{a^2 - b^2} \left| \frac{-2\sin(\theta - \phi)}{\sin(\theta + \phi)\sin(\theta - \phi)} \right| \quad \text{as } a^2 > b^2$$

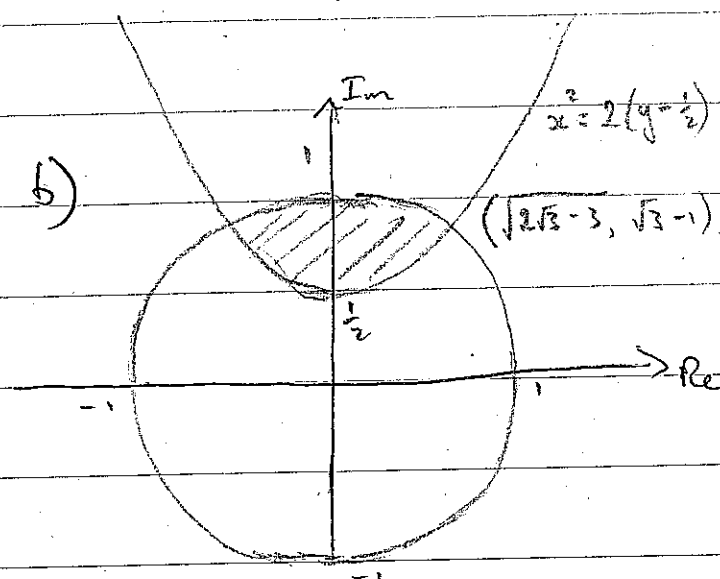
$$= \frac{2ab}{a^2 - b^2} \left| \operatorname{cosec}(\theta + \phi) \right|$$

Q3 a) $(3-i)z + (2+3i) = \overline{8+7i} \Rightarrow z = \frac{8-7i - (2+3i)}{3-i}$

$$= \frac{(6-10i)(3+i)}{9+1}$$

$$= \frac{14-12i}{10}$$

b)



$$(\sqrt{2}\sqrt{3}-3, \sqrt{3}-1)$$

| circle

| parabola

| shading

c) $|- \sqrt{3} + i| = 2$; $\arg(-\sqrt{3} + i) = 150^\circ \text{ or } \frac{5\pi}{6}$

$$(-\sqrt{3} + i)^7 = 2^7 \operatorname{cis} \frac{35\pi}{6} \quad [\text{de Moivre}]$$

$$= 2^7 \operatorname{cis} \left(-\frac{\pi}{6}\right) = 2^6 (\sqrt{3} - i)$$

d) (i) Sum of roots $= -3 + 2i \Rightarrow$ other root $= 1$

(ii) Product of roots $= k = -4 + 2i$

e) AB $+ zw$

OB $= zw^2$ or $z(w+1)$ (or $-z\bar{w}$)

f) $|z^2 + (\operatorname{Re} z)^2| = 4$

$$|2x^2 - y^2 + 2ixy| = 4$$

$$(2x^2 - y^2)^2 + 4x^2y^2 = 16$$

$$4x^4 + y^4 = 16$$

Progress |

Q4 a) $T_1 = 1$, $T_2 = 4$ and $T_n = 6T_{n-2} + T_{n-1}$, $n \geq 2$

RTP $T_n = \frac{2 \times 3^n - (-2)^{n+1}}{5}$ for $n \geq 1$

When $n=1$ $RHS = \frac{2 \times 3 - 1}{5} = 1$

When $n=2$ $RHS = \frac{2 \times 9 + 2}{5} = 4$

$\therefore$ The proposition is true for $n=1$ and $n=2$

Assume that $T_k = \frac{2 \times 3^k - (-2)^{k+1}}{5}$ and $T_{k+1} = \frac{2 \times 3^{k+1} - (-2)^{k+2}}{5}$ for $k \geq 1$

Then $T_{k+2} = 6 \times \frac{2 \times 3^k - (-2)^{k+1}}{5} + \frac{2 \times 3^{k+1} - (-2)^{k+2}}{5}$
 $= \frac{1}{5} \left\{ 2 \times (6 \times 3^k + 3^{k+1}) - 6 \times (-2)^{k+1} - (-2)^{k+2} \right\}$
 $= \frac{1}{5} \left\{ 2 \times 9 \times 3^k - 4 \times (-2)^{k+1} \right\}$
 $= \frac{2 \times 3^{k+2} - (-2)^{k+2}}{5}$ as required

Hence the proposition is proved true by induction.

b) (i) $(\sqrt{x} - \sqrt{y})^2 \geq 0$ for $x, y > 0$

$\therefore x - 2\sqrt{xy} + y \geq 0$

$\therefore x + y \geq 2\sqrt{xy}$

(ii) Let $x = 2^n$ and $y = 2^m$ Clearly $x, y > 0$ for real n, m

$= 2^n + 2^m \geq 2\sqrt{2^{n+m}}$
 $= 2^{\frac{n+m+2}{2}}$

$\therefore \log_2 (2^n + 2^m) \geq \frac{n+m+2}{2}$

noting that $y = \log_2 x$ is an increasing function
 $\therefore$ Inequality maintains its direction.

(c) (i) RTP $(a+1)^n - a^n > n a^{n-1}$ for $a > 0$, $n \geq 2$

For $n=2$ LHS $= a^2 + 2a + 1 - a^2$

$= 2a + 1$

RHS $= 2a$ $\therefore$ True for $n=2$

Assume that $(a+1)^k - a^k > k a^{k-1}$ for an integer $k \geq 2$

Now $(a+1)^{k+1} - a^{k+1} = a \cdot [(a+1)^k - a^k] + (a+1)^k$
 $> k a^k + (a+1)^k$ by assumption
 $> k a^k + a^k$
 $= (k+1) a^k$

Hence it is true for $n=k+1$ if it is true for $n=k$.

It is true for $n=2$ $\therefore$ True for all $n \geq 2$ by induction.

(ii) $n a^{n-1} < (a+1)^n - a^n < n(a+1)^{n-1}$ from above plus given result.
 $a^{n-1} < \frac{(a+1)^n - a^n}{n} < (a+1)^{n-1}$
 $a < \left(\frac{(a+1)^n - a^n}{n} \right)^{\frac{1}{n-1}} < a+1$ since $a > 0$

(iii) (A) $f(x) = [(a+1)^n - a^n]x + a^n - [(a+1)^n - a^n] - x^n$

$f'(x) = [(a+1)^n - a^n] - n x^{n-1}$

$f''(x) = -n(n-1)x^{n-2} < 0$ for all $n \geq 2$ and $x > 0$ $\therefore$ Concave down

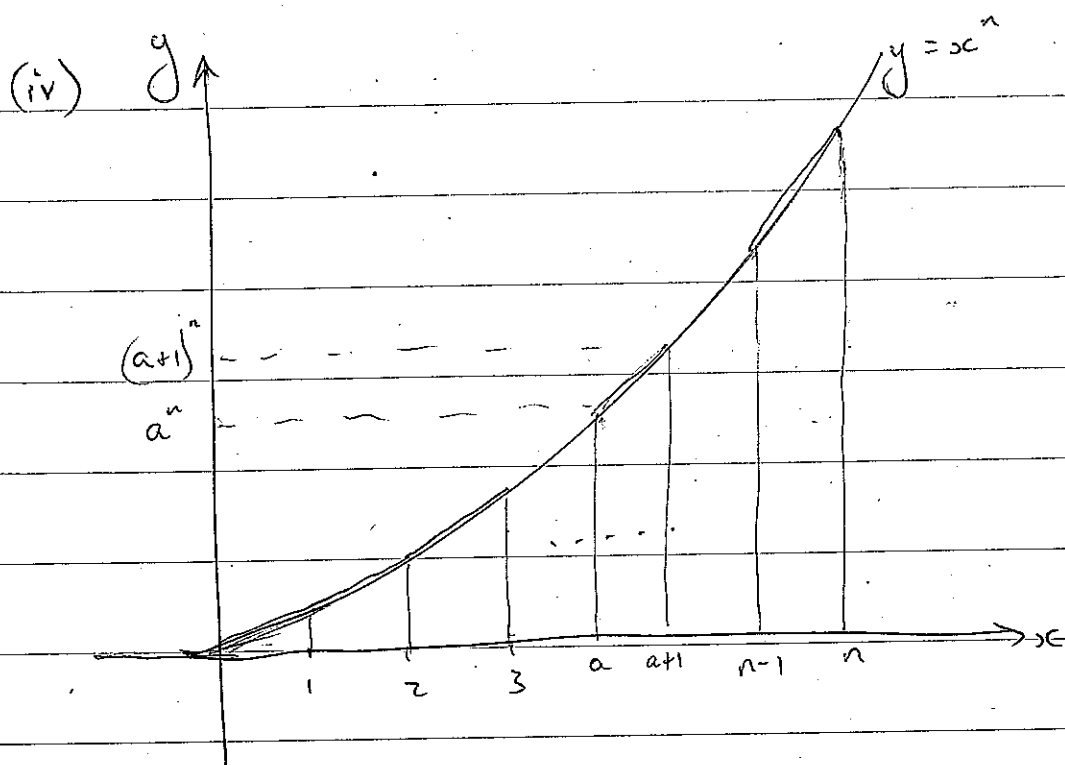
$f'(x) = 0 \Rightarrow x = \left[\frac{(a+1)^n - a^n}{n} \right]^{\frac{1}{n-1}}$

$y = f(x)$ is concave down $\therefore$ Maximum. From part (ii) $a < x < a+1$

(B) $f(a) = f(a+1) = 0$ and there is exactly one

stationary point in $a < x < a+1$, and that

point is a maximum $\therefore f(x) > 0$ for $a < x < a+1$



Approximating the curve with straight line segments from (a, a^n) to $(a+1, (a+1)^n)$ for $a = 0, 1, 2, \dots, (n-1)$ we form n trapeziums. From part (iii) (β) we know these intervals lie above $y = x^n$ for $a < x < a+1; n \geq 2$.
 $\therefore$ The exact area $<$ The sum of the areas of the trapeziums

$$\therefore \int_0^n x^n dx < \frac{1}{2} \times 1 \times (0^n + 1^n) + \frac{1}{2} \times 1 \times (1^n + 2^n) + \frac{1}{2} \times 1 \times (2^n + 3^n) + \dots + \frac{1}{2} \times 1 \times ((n-2)^n + (n-1)^n) + \frac{1}{2} \times 1 \times ((n-1)^n + n^n)$$

$$\therefore \frac{n^{n+1}}{n+1} < \frac{1}{2} \times 0^n + 1^n + 2^n + 3^n + \dots + (n-1)^n + \frac{1}{2} \times n^n$$

$$\frac{n^{n+1}}{n+1} < 1 + 2^n + 3^n + \dots + (n-1)^n + \frac{n^n}{2}$$

for $n \geq 2$